Practical 3
Sudo mysql
Create databse college;
Use college;

```
CREATE TABLE Students (
    student_id INT PRIMARY KEY,
    name VARCHAR(30),
    city VARCHAR(30));

CREATE TABLE Enrollments (
    enroll_id INT PRIMARY KEY,
    student_id INT,
    course VARCHAR(30),
    marks INT,
    FOREIGN KEY (student_id) REFERENCES Students(student_id));

INSERT INTO Students VALUES
(1, 'Aarav', 'Pune'),
(2, 'Priya', 'Mumbai'),
(3, 'Pooja', 'Pune'),
(4, 'Rahul', 'Nashik'),
(5, 'Pranav', 'Mumbai'),
(6, 'Sneha', 'Pune'),
(7, 'Vikram', 'Delhi'),
(8, 'Payal', 'Mumbai'),
(9, 'Neha', 'Nagpur'),
(10, 'Karan', 'Pune');

INSERT INTO Enrollments VALUES
(101, 1, 'DBMS', 85),
(102, 2, 'Java', 78),
(103, 3, 'DBMS', 92),
(104, 4, 'Python', 67),
(105, 5, 'Java', 88),
(106, 6, 'DBMS', 74),
(107, 7, 'Python', 95),
(108, 8, 'Java', 81),
(109, 9, 'DBMS', 59),
(110, 1, 'Python', 90);
```

1. select s.name,e.course from Students s inner join Enrollments e on s.student_id =e.student_id;

2. select s.name,e.course,e.marks,s.student_id from Students s left outer join Enrollments e on s.student_id=e.student_id;

3. select s.name,e.course from Students s inner join Enrollments e on s.student_id=e.student_id where e.course='DBMS';

4. SELECT e.* FROM Enrollments e LEFT JOIN Students s ON e.student_id = s.student_id WHERE s.student_id IS NULL;

5. SELECT s.name, e.course, e.marks FROM Students s INNER JOIN Enrollments e ON s.student_id = e.student_id WHERE e.marks < (SELECT AVG(marks) FROM Enrollments);

6. SELECT s.student_id, s.name, e.marks FROM Students s INNER JOIN Enrollments e ON s.student_id = e.student_id WHERE e.marks = (SELECT MAX(marks) FROM Enrollments);

7. select * from Students where city ='Pune';

8. select s.student_id,s.name,e.course from Students s inner join Enrollments e on s.student_id=e.student_id where e.course='Python';

9. create view course_average_view as select course,avg(marks) as avg_marks from Enrollments group by course;

   SELECT * FROM course_average_view;

10. CREATE INDEX idx_student_city ON Students(city);
    CREATE UNIQUE INDEX idx_unique_student_id ON Students(student_id);

SHOW INDEX FROM Students;